

1. Project Structure

```
university-dept/  
├── index.html  
├── styles.css  
├── Dockerfile  
├── Jenkinsfile  
└── k8s-deployment.yaml
```

2. index.html

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>University Department</title>  
  <link rel="stylesheet" href="styles.css">  
</head>  
<body>  
  
<header>  
  <h1>ABC University</h1>  
  <p>Top-tier institution focused on innovation and research</p>  
  <h3>Head: Dr. Kumar (HOD)</h3>  
</header>  
  
<section>  
  <h2>About Department</h2>  
  <p>Computer Science Department focuses on AI, Data Science, and Software  
Engineering.</p>  
</section>  
  
<section>  
  <h2>Courses</h2>  
  <ul>  
    <li>Data Structures - Core programming concepts</li>  
    <li>Machine Learning - AI-based systems</li>  
    <li>DBMS - Database design</li>  
  </ul>  
</section>
```

```
<section>
  <h2>Faculty</h2>
  <p>Dr. Ravi - AI Expert</p>
  <p>Dr. Meena - DBMS Specialist</p>
</section>

<section>
  <h2>Contact</h2>
  <p>Address: Katpadi, Tamil Nadu</p>
  <p>Phone: 9876543210</p>
  <p>Email: cs@abcuniv.edu</p>
</section>

<footer>
  <p>© 2026 ABC University</p>
</footer>

</body>
</html>
```

3. styles.css

```
body{
  font-family: Arial;
  margin: 20px;
}

header, footer {
  background: #333;
  color: white;
  padding: 10px;
}

section {
  margin: 20px 0;
}
```

4. Git Setup

```
git init
git add .
git commit -m "Initial commit - University Webpage"
```

Create repo on GitHub

```
git remote add origin https://github.com/your-username/university-dept.git
git branch -M main
git push -u origin main
```

5. Clone + Modify

```
git clone https://github.com/your-username/university-dept.git
cd university-dept
code .
```

Make small change → then:

```
git add .
git commit -m "Updated content"
git push
```

6. Dockerfile

```
FROM nginx:latest
```

```
COPY . /usr/share/nginx/html
```

```
EXPOSE 80
```

7. Build & Run Docker

```
docker build -t university-web .
docker run -d -p 8080:80 university-web
```

👉 Open: <http://localhost:8080>

8. k8s-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
```

```
metadata:
  name: university-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: university
  template:
    metadata:
      labels:
        app: university
    spec:
      containers:
        - name: university-container
          image: university-web
          ports:
            - containerPort: 80
```

```
apiVersion: v1
kind: Service
metadata:
  name: university-service
spec:
  type: NodePort
  selector:
    app: university
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

9. Apply Kubernetes

```
kubectl apply -f k8s-deployment.yaml
kubectl get pods
kubectl get services
```

👉 Access via NodePort (30007)

10. Jenkinsfile

```

pipeline {
  agent any

  stages {

    stage('Clone') {
      steps {
        git 'https://github.com/your-username/university-dept.git'
      }
    }

    stage('Build Docker') {
      steps {
        bat 'docker build -t university-web .'
      }
    }

    stage('Run Container') {
      steps {
        bat 'docker run -d -p 8080:80 university-web'
      }
    }

    stage('Deploy to K8s') {
      steps {
        bat 'kubectl apply -f k8s-deployment.yaml'
      }
    }
  }
}

```

👉 Linux → replace bat with sh

11. Jenkins Setup

In Jenkins:

- New Item → Pipeline
- Paste Jenkinsfile
- Click **Build Now**

12. Verify in Docker Desktop

- Open Docker Desktop
- Check running container
- Check Kubernetes pods